Four homologies in a projective plane

A recovery of the odd-characteristic rank-four classification in L027

Let F have odd characteristic p. We classify all rank-four string C-markings in $\mathrm{PSL}_3(F)$, including the degenerate case, and obtain the directly regular alternatives needed in Section 6 of Leemans–Vandenschrick. The normal forms originate in their Section 5.3. The proof below uses the actual translation kernels and an acquired projective-line classification, without first classifying all rank-three subgroups of $\mathrm{PSL}_3(F)$.

We retain the usual string end-intersection criterion, quoted as Lemma 2.1 of the source, and the classical identification $\mathrm{SO}_3(F) \cong \mathrm{PGL}_2(F)$. We also acquire the previously proved Topic E classification: a rank-four string C-marking of $\mathrm{PSL}_2(K)$ or $\mathrm{PGL}_2(K)$ is exactly the $\mathrm{PSL}_2(11)$ marking of type $[3,5,3]$, the $\mathrm{PSL}_2(19)$ marking of type $[5,3,5]$, or the $\mathrm{PGL}_2(5)$ marking of type $[3,3,3]$. Its finite presentations and complete intersection checks remain in that acquired component. Dickson's subgroup theorem, in its actual natural-subfield form, is the only classification input used in passing to those groups.

The diagonal outer pair

Lemma 1. *A string C-marking in $\mathrm{PSL}_3(F)$ has rank at most four. Every rank-four marking has, after a projective change of coordinates, unique order-two lifts of the form*

$$r_0 = \mathrm{diag}(1,-1,-1), \qquad r_3 = \mathrm{diag}(-1,-1,1),$$
$$r_1 = \begin{pmatrix} a & b & 0 \\ c & -a & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad r_2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -d & f \\ 0 & e & d \end{pmatrix}, \qquad bc = 1-a^2, \quad ef = 1-d^2.$$

Here $a, d \notin \{0,-1\}$. If either is one, exactly one of its two off-diagonal parameters is zero.

Proof. The scalar centre has odd order dividing three, so projective involutions lift uniquely to involutions of determinant one. Projectively commuting lifts commute: a scalar commutator ζ would have $\zeta^2 = 1$. In rank at least five, the first, third and fifth generators give an independent C_2^3. Their lifts are simultaneously diagonalizable, whereas the determinant-one diagonal sign group has order four. This contradicts independence.

At rank four, simultaneously diagonalize the independent outer pair as r_0, r_3 above. The involution r_1 preserves both eigenspaces of r_3. It acts as -1 on $\langle e_3 \rangle$, since otherwise its one-dimensional $+1$ eigenspace would be that point and $r_1 = r_3$. Its remaining block has trace zero and determinant -1. The same argument applies to r_2, giving the displayed forms. If $a = 0$, then $(r_0 r_1)^2 = r_3$, contrary to IP. If $a = -1$, then $(r_0 r_1)^p = r_3$, again impossible. If $a = 1$ and $b = c = 0$, then $r_1 = r_0$. Reversal gives the same assertions for d, e, f. □

The conic alternative

Lemma 2. *A dihedral group has no nondegenerate rank-three string C-marking.*

Proof. All three generators must be reflections: a central involution would commute with its neighbour. Write the commuting distinct ends as r and rz, where z is the central involution.

If s is the middle reflection, then $(srz)^2 = (rs)^{-2}$. Since $|rs| > 2$, this is a nonidentity rotation in both dihedral ends and not in $\langle s \rangle$, contradicting IP. □

Lemma 3. *If all four off-diagonal parameters b, c, e, f are nonzero, the marking is one of the three acquired projective-line markings. The ambient characteristics are respectively* $11, 19, 5$.

Proof. Direct multiplication shows that the four generators preserve the nonsingular diagonal quadratic form

$$Q = (c/b)x_1^2 + x_2^2 + (f/e)x_3^2.$$

The resulting orthogonal group has trivial intersection with the scalar centre of $\mathrm{SL}_3(F)$, since a scalar preserving Q has square one. Thus the marked group embeds faithfully in $\mathrm{PGL}_2(F)$. Its outer labels exceed two by Lemma 1, and the middle generators do not commute: the $(1,3)$ entry of $r_1 r_2$ is $bf \neq 0$, whereas that of $r_2 r_1$ is zero. The marking is nondegenerate.

We apply Dickson's subgroup alternatives to this actual subgroup. Cyclic groups and the involution-generated subgroup V_4 of A_4 are excluded. In an odd-characteristic affine group every involution has the form $x \mapsto -x + v$, and two distinct such involutions do not commute; the commuting first and third generators exclude that case. The dihedral case is excluded by Lemma 2 applied to an end. If the whole group were S_4 or A_5, either proper rank-three end would lie in an A_4, a dihedral subgroup or an abelian subgroup, by their elementary proper-subgroup lists; none admits a nondegenerate rank-three marking. IP makes these ends proper, so these two cases are excluded as well.

The group is therefore a natural subfield $\mathrm{PSL}_2(K)$ or $\mathrm{PGL}_2(K)$. The acquired Topic E theorem gives the three marked possibilities. The orders and simplicity distinguish them: $\mathrm{PSL}_2(11)$ and $\mathrm{PSL}_2(19)$ are simple, while $\mathrm{PGL}_2(5)$ has an index-two subgroup. Their natural subfield embeddings consequently have characteristic $11, 19, 5$, respectively. Conversely their standard three-dimensional orthogonal representations embed them in $\mathrm{PSL}_3(F)$ in those characteristics, with the same actual markings. □

The affine alternatives

Duality transposes each lift, interchanging b, c and also e, f. Reversal, followed by interchange of the first and third coordinates, interchanges the two pairs. These operations reduce all remaining cases to $b = 0$, with either no further zero, $e = 0$, or $f = 0$. The nonzero parameter c may be scaled to one; when $d = 1$, the other nonzero off-diagonal parameter can also be scaled to one. These are changes of basis, not changes to the ordered generated group.

Lemma 4. *The case $b = 0$ and $ef \neq 0$ cannot have IP.*

Proof. Here $a = 1$, $c \neq 0$, and $d \notin \{0, 1, -1\}$. In block form with the invariant plane $\langle e_2, e_3 \rangle$, write

$$r_1 = \begin{pmatrix} 1 & 0 \\ ce_1 & -I_2 \end{pmatrix}, \quad r_2 = \mathrm{diag}(-1, P), \quad r_3 = \mathrm{diag}(-1, Q_0).$$

Here

$$P = \begin{pmatrix} -d & f \\ e & d \end{pmatrix}, \qquad Q_0 = \mathrm{diag}(-1, 1).$$

Let $T(v) = \left(\begin{smallmatrix} 1 & 0 \\ v & I_2 \end{smallmatrix}\right)$. The actual subgroup $B = \langle r_1, r_2, r_3 \rangle$ contains $(r_1 r_2)^2 = T((1-d)c, ec)$. Its intersection with the additive translation group is invariant under $-P$ and $-Q_0$. The latter is the sign change of the second coordinate. Adding and subtracting its image and dividing by two projects onto each coordinate. Hence this intersection contains $T((1-d)c, 0)$. Applying $-P$ and projecting again multiplies that coordinate by d. Since $\mathbb{F}_p[d]$ is a field and $1 - d \neq 0$, its polynomial span contains $T(c, 0)$. But $r_1 = T(c, 0) r_0$, so $r_0 \in B$, contrary to IP. □

Theorem 5. *The two remaining affine cases are as follows.*

1. *If $b = e = 0$, the marking has type $[p, 2, p]$ and group $D_{2p} \times D_{2p}$.*
2. *If $b = f = 0$, the marking has type $[p, 2p, p]$ and group $\mathrm{Heis}(p) \rtimes C_2^2$, where the Heisenberg group consists of all lower unitriangular 3×3 matrices over $\mathbb{F}_p$.*

Each displayed normal form has full IP and is directly regular.

Proof. In the first case, after scaling $c = f = 1$, the two cyclic translation groups $\langle I + E_{21} \rangle$ and $\langle I + E_{23} \rangle$ commute. The diagonal r_0 inverts the first and fixes the second; r_3 does the reverse. This is the actual direct product of two dihedral groups, with its two rank-two markings placed consecutively. The type and all subset intersections follow from that direct product.

In the second case set $c = e = 1$. Put

$$h(x, y, z) = \begin{pmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ z & y & 1 \end{pmatrix}, \quad D_0 = r_0, \quad D_3 = r_3.$$

The root groups $h(x, 0, 0)$ and $h(0, y, 0)$, generated respectively by $r_0 r_1$ and $r_2 r_3$, have commutators giving all $h(0, 0, z)$, with $x, y, z \in \mathbb{F}_p$. The two diagonal generators normalize this Heisenberg group and intersect it trivially. The full group thus consists uniquely of $h(x, y, z) D_0^i D_3^j$, with $i, j \in \{0, 1\}$, and has order $4p^3$. Here

$$r_1 = h(1, 0, 0) D_0, \qquad r_2 = h(0, -1, 0) D_3.$$

The first actual end $A = \langle r_0, r_1, r_2 \rangle$ consists of these normal forms with $y = -j$, and the second $B = \langle r_1, r_2, r_3 \rangle$ consists of those with $x = i$. For example A contains the full x, z translation plane (conjugate the x root group by r_2), and its complement is the commuting pair r_0, r_2. This proves both inclusions in its displayed description; the other end follows in the same way.

Their intersection has $x = i, y = -j$ and arbitrary z, hence order $4p$. Direct multiplication gives $(r_1 r_2)^2 = h(0, 0, 1)$ and shows that $r_1 r_2$ has order $2p$. Consequently $A \cap B = \langle r_1, r_2 \rangle$. Within A, the subgroup $\langle r_0, r_1 \rangle$ has $j = y = z = 0$; intersecting it with the preceding description of $\langle r_1, r_2 \rangle$ leaves exactly $\langle r_1 \rangle$. Reversal gives the other rank-three test. The end criterion proves full IP. The outer products have order p, giving type $[p, 2p, p]$.

Finally, in both cases the map to C_2 which sends a diagonal $D_0^i D_3^j$ to $i + j$ sends all four generators to its nonidentity element. Its kernel is exactly the even-word subgroup, proving direct regularity. □

Classification and source qualifications

Combining the lemmas gives exactly the source's Theorem 5.3: the degenerate direct product, the Heisenberg extension, and the three conic markings. The nondegenerate directly regular possibilities are only the Heisenberg marking of type $[p, 2p, p]$ and the simplex marking of type $[3, 3, 3]$ in characteristic five. The two simple conic groups have no index-two subgroup and are not directly regular. The simplex marking is directly regular by the usual sign map on S_5.

The source's physical page 17 names D_{2p} in the two-zero direct product case; its Theorem 5.3 correctly gives D_{2p}^2. The actual matrix decomposition above settles this discrepancy. Its two-point description also requires allowing duality: our chosen form fixes two lines, and its dual fixes two points. The restriction "$q = 5$" in the rank-six discussion must mean characteristic five for this subgroup alternative, since the same natural subfield marking embeds for every extension of $\mathbb{F}_5$. This is not yet a proof of the rank-six exclusion. The source's Table 1 parameter calculations are replaced here by the acquired complete marking classification;

no new finite census is charged to that previously recovered result. The broader rank-three classifications in Theorems 5.1–5.2 remain a separate source-contract item.